

Prompt Engineering PDF for the Exam

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1 Part 1: Prompt Comparison Analysis (30 points) Task

I choose healthcare for my domain

Here's the Version A

I write to AI (CHATGPT) this
Tell me about healthcare?



Definition

Healthcare is the **prevention, diagnosis, treatment, and management** of illness, injury, and other physical or mental impairments in people. It's provided by professionals such as doctors, nurses, pharmacists, dentists, therapists, and public health workers.

In return i got this it's a basic and vague prompt

Here's the Version B

I asked to CHATGPT this

Explain asthma in simple terms, describing what it is, its stages or severity levels, and possible treatments including medications and lifestyle management.

and this is what i got

Structured Output: Asthma Explained Simply	3. Treatments
1. What Asthma Is (Simple Terms) - Asthma is a long-term lung condition. - Your airways (the tubes that carry air in and out) become sensitive, tight, and sometimes swollen. - This makes it hard to breathe, especially during an asthma attack.	A. Medications 1. Quick-Relief Medicines (Rescue) - Short-acting bronchodilators (e.g., albuterol). - Used during an asthma attack. - They relax and open the airways fast. 2. Long-Term Control Medicines - Inhaled corticosteroids (reduce swelling in airways). - Long-acting bronchodilators (LABA). - Leukotriene modifiers (tablet form). - Taken daily to prevent attacks. 3. Severe Asthma Treatments - Biologic injections (for people who don't respond to normal meds). - Oral steroids (short-term during strong attacks).
2. Stages / Severity Levels a. Intermittent Asthma - Symptoms happen once in a while. - Breathing is normal between episodes. b. Mild Persistent Asthma - Symptoms appear more than twice a week, but not daily. - Slight problems with daily activities. c. Moderate Persistent Asthma - Symptoms happen every day. - Inhaler needed often. - Sleep and activity may be affected. d. Severe Persistent Asthma - Symptoms happen throughout the day. - Breathing problems are frequent. - Normal daily activities become difficult.	4. Lifestyle Management - Avoid triggers (dust, smoke, pollen, cold air, strong smells). - Use your inhaler correctly (good technique is important). - Exercise regularly, but warm up first. - Keep home clean to reduce dust/allergens. - Monitor symptoms with an action plan from your doctor. - Stay healthy: good sleep, hydration, and stress control.

it's structured and useful

I used Structured Output because I wanted to learn more about asthma such as stages possible treatments etc I think structured output is to best way to learn about this topic because there's so many things to learn about asthma

When we're comparing Version A and Version B, Version A is a basic response with little information about healthcare. As you said in class, a good answer is given to the person who asks a good question. It's just a basic question with a basic answer.

When comparing it to Version B, I made it more specific because there are many things about healthcare. I chose asthma, and I also wanted to learn about its stages and possible treatments.

Here are three concrete improvements in Version B:

- I wanted specific information about a problem, so I chose asthma.
- I also wanted its stages and possible treatments, so I could not only learn about asthma but also other things related to it.
- It's longer and more useful for the user.

The improved prompt worked better because I got the information I wanted. Basically, I asked a good question and in return, I got a really good answer. In Version A, I asked a basic and weak question, so I received a short and simple answer. But in Version B, I asked a detailed and clear question, and in return, I got a better, more detailed, and longer answer. Most importantly, I learned something that could be useful for me.

2 Part 2: Practical Prompt Design

I chose the CO-STAR Framework to design a social media content calendar for one month. This framework is effective because it clearly separates the context, objective, style, tone, audience, and response format. For content creation tasks, these elements ensure the AI understands the brand's purpose, voice, and audience expectations. By using CO-STAR, I can guide the AI to produce consistent, engaging, and brand-aligned social media posts suitable for small business marketing goals.

C (Context): I'm a digital marketing manager for a small fitness brand that focuses on home workouts and nutrition tips.

O (Objective): Create a one-month social media content calendar to increase engagement and brand awareness.

S (Style): Posts should be short, visually engaging, and easily shareable. Include a mix of educational, motivational, and promotional posts.

T (Tone): Friendly, energetic, and motivating.

A (Audience): Busy adults aged 20–40 who want to stay fit at home.

R (Response): Provide a 30-day content calendar in a table with columns for Date, Post Type, Topic/Caption, Platform (Instagram, TikTok, or Facebook), and Goal (e.g. engagement, traffic, sales).

Date	Post Type	Topic/Caption	Platform	Goal
Nov 1	Motivational	"Start your week strong 🍊 15-min morning stretch for energy!"	Instagram	Engagement
Nov 2	Educational	"5 quick nutrition tips to fuel your home workouts 🍎 🥑 "	Facebook	Traffic
Nov 3	Promotional	"New resistance bands now available! Upgrade your home gym today 🏠 "	Instagram	Sales
Nov 4	Interactive	"Poll: Which workout do you prefer? Cardio or Strength? Vote now!"	Instagram	Engagement
Nov 5	Motivational	"Consistency > perfection. Keep moving! 💪 "	TikTok	Engagement
Nov 6	Educational	"3 exercises you can do in your living room 🏠 "	Facebook	Traffic
Nov 7	Promotional	"Limited-time bundle: Home workout + nutrition guide 📖 "	Instagram	Sales
...	...	...	...	...
Nov 30	Motivational	"Reflect on your progress this month — small wins matter 🌟 "	TikTok	Engagement

Framework adherence highlights:

C (Context): Posts revolve around home workouts and nutrition tips.

O (Objective): Calendar designed to boost engagement, traffic, and sales.

S (Style): Short captions, hashtags, emoji use, visually engaging formats like Reels, Stories, and polls.

T (Tone): Friendly, energetic, motivating tone is maintained throughout.

A (Audience): Targets busy adults with quick, easy-to-digest content and actionable tips.

R (Response): Delivered as a structured table covering the 30-day plan (detailed start + last day).

Weaknesses / Areas for Improvement

Lack of Visual & Hashtag Guidance:

The calendar lists captions and topics but doesn't suggest visuals, video styles, or trending hashtags/sounds, which are crucial for making posts engaging and shareable on social media.

Limited Platform-Specific Optimization:

Posts are repeated across Instagram, TikTok, and Facebook without tailoring content for each platform's strengths. TikTok favors short vertical videos with trending audio, while Instagram thrives on high-quality images, reels, and hashtags. Without platform-specific formatting, engagement may be lower.

Date	Post Type	Topic / Caption	Platform	Visual / Format	Hashtags / Sounds	Goal	
Day 1	Motivational	"New month, new goals! 🍌 What's your fitness goal?"	Instagram	Branded graphic with month goals	#FitnessGoals #HomeWorkout	Engagement	
Day 2	Educational	5 quick home stretches to start your day 🧘	TikTok	Vertical video showing stretches	Trending upbeat sound	Engagement	
Day 3	Promotional	Limited-time offer: 1-week meal plan \$9.99! 🍏	Facebook	Carousel with meal plan visuals	#HealthyEating #HomeFitness	Sales	
Day 4	Workout Tip	10-min HIIT for busy mornings 🏃	Instagram	Reel showing workout moves	#HIITWorkout #FitLife	Traffic	
Day 5	Motivational	Transformation Tuesday: Share your progress pics! ✨	Instagram	User-generated content collage	#ProgressPics #Motivation	Engagement	
Day 6	Educational	Benefits of drinking water before workouts 💧	TikTok	Short video with animations	Trending wellness sound	Engagement	
Day 7	Interactive	Poll: Cardio or Strength Training—your pick?	Instagram	Story poll with GIFs	#FitnessPoll #WorkoutChoice	Engagement	
Day 30	Wrap-up / Motivational	Celebrate your wins this month! 🎉 Share your next goal	Instagram	Carousel of user posts & motivational quote	#MonthlyWins #HomeFitness	Engagement	

The refinement adds actionable guidance for visuals, hashtags, and platform optimization while keeping the original tone, audience, and content mix. This makes the calendar more practical, engaging, and ready for real-world posting.

Critical Reflection: Prompt Engineering Journey

a. Skill Progression

I think my understanding of prompt engineering has changed a lot over time shifting from just seeing it like a simple dialog with a chatbot to more like programming with words at first I was only writing short one-step questions and not really getting good results but then I realize how important structure and clear outputs are. my aha moment was when I learn about using a kind of structural contract like forcing the answer into a JSON or table this made everything more stable and useful because it stop being random text and become more like data that can be used later

the hardest part for me was about how models can make up information they sound confident but sometimes wrong I overcome this by trying something called reflection prompting and also by adding some governance ideas to make the model check itself before finishing I think this was very useful because it made the answers less fake

b. Practical Application

In my work in financial marketing and content creation I can use prompt engineering to make automatic compliance check drafts I want to make product descriptions that sound good for marketing but also fit the legal rules from our company and not cause problems

I would use a pattern stack and something like a two-pass generation first for the marketing part I ask it to act as a persuasive and friendly writer and focus on benefits then in the second pass I add a part with clear separators and ask for a neutral legal disclaimer text based on the rules we already have

the main issue here is tone bleed where the friendly tone leaks into the legal part and makes it sound too soft I fix this by adding clear marks between sections and reminding the model to reset tone and role before starting the disclaimer part

c. Limitations & Ethics

When I prompt I always try to remember two big limits one is hallucination when it makes something up that looks real but isn't true the other one is that it can forget or cut long text because of its context size so when dealing with long financial rules some parts may go missing or be wrong

Another thing I care about is ethics and plagiarism I don't want AI to copy from somewhere else or use trademarked things the content must be original I always ask it to show where it gets facts from and after that I check everything myself to make sure the info and tone are right before using it anywhere

Goals for the Second Half

Well because of technical difficulties I couldn't joined to classes which is unlucky for me
I'd like to learn more things about complex problem solving I also hope to improve my
ability to debug and test prompts more efficiently

Just used Chatgpt for outputs also to fix my grammar errors

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